

Exponential-size rational equal-weight interval quadrature

An alternative local proof; global inputs are classical

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Attribution update. The stronger upper bound Cr^32^r is already a consequence of Arkhipov’s classical Hilbert–Kamke results; see `arkhipov_rational_corollary.tex`. The argument below is retained as an independent proof of local solubility. It is not a claim that exponential rational upper bounds are historically new.

Theorem 1. *There is an absolute constant C such that, for every $r \geq 2$ and every integer $K \geq Cr^64^r$, there are rational numbers $x_1, \dots, x_K \in (0, 1)$ satisfying*

$$\frac{1}{K} \sum_{i=1}^K x_i^j = \frac{1}{j+1} \quad (0 \leq j \leq r).$$

In particular the minimum rational equal-weight quadrature size has $\log K = \Theta(r)$, in conjunction with the independently proved p -adic exponential lower bound.

The analytic ingredient is Cui–Xia–Xiang, Theorem 5.2(i),(iv),(v) [1], used below with its stated threshold $K \geq 3r^22^r - r$. We give the quantitative local-solubility argument in full. No bound on the common denominator of the x_i is claimed.

1 A quantitative inverse-function lemma

Write v_p for the valuation normalized by $v_p(p) = 1$; valuations of vectors and matrices mean the minimum valuation of their entries.

Lemma 2. *Let $F : \mathbb{Z}_p^r \rightarrow \mathbb{Z}_p^r$ be a polynomial map with integral coefficients, $u \in \mathbb{Z}_p^r$, and $A = DF(u)$ invertible over $\mathbb{Q}_p$. Suppose $v_p(A^{-1}) \geq -b$, where $b \geq 0$ is an integer. If $v_p(F(u)) \geq 2b + 1$, then F has a zero $u + h$ with $h \in p^{b+1}\mathbb{Z}_p^r$. Its Jacobian is invertible over $\mathbb{Q}_p$.*

Proof. Write $F(u+h) = F(u) + Ah + R(h)$, where R has integral coefficients and no terms of degrees zero or one. On the ball $B = p^{b+1}\mathbb{Z}_p^r$, the map

$$\Phi(h) = -A^{-1}F(u) - A^{-1}R(h)$$

maps B to itself. Moreover, for $h, h' \in B$, every monomial of degree at least two gives

$$v_p(R(h) - R(h')) \geq b + 1 + v_p(h - h'),$$

so Φ is a strict contraction, increasing the valuation of a difference by at least one. Completeness gives its fixed point. Finally $DF(u+h) = A + E$ with $v_p(E) \geq b + 1$, whence $A^{-1}DF(u+h) = I + A^{-1}E$ is congruent to I modulo p . $\square$

2 Small primes: a smooth homogeneous zero

Lemma 3. *For $r \geq 2$ and a prime p , put*

$$b_p = v_p((r-1)!) + \lfloor \log_p r \rfloor, \quad a_p = 2b_p + 2, \quad M_p = p^{a_p}.$$

For every $K \geq M_p$, the homogeneous system

$$\sum_{i=1}^K z_i^j = 0 \quad (1 \leq j \leq r) \tag{1}$$

has a nonsingular solution in $\mathbb{Z}_p^K$.

Proof. Start with the M_p integers $0, 1, \dots, M_p - 1$, and append zeros if $K > M_p$. For every $j \geq 1$ and $a \geq 1$,

$$v_p \left(\sum_{x=0}^{p^a-1} x^j \right) \geq a - 1. \tag{2}$$

Indeed the assertion at $a = 1$ is immediate; writing the residues modulo p^{a+1} as $x + tp^a$ shows that the sum equals p times the previous sum modulo p^a , proving the inductive step.

Only the first r coordinates, initially $0, 1, \dots, r-1$, are varied. They exist since $a_p \geq 2\lfloor \log_p r \rfloor + 2$ implies $M_p \geq r$. Their Jacobian has entries $A_{j,i} = ji^{j-1}$, $1 \leq j \leq r$, $0 \leq i \leq r-1$. To bound its inverse, use the Lagrange polynomials

$$L_i(X) = \prod_{\substack{0 \leq h \leq r-1 \\ h \neq i}} \frac{X - h}{i - h}.$$

Every coefficient has denominator dividing $i!(r-1-i)!$, which divides $(r-1)!$. Thus the inverse of the Vandermonde matrix $(i^{j-1})_{j,i}$ has valuation at least $-v_p((r-1)!)$. The additional diagonal matrix with entries $1, \dots, r$ costs at most $\lfloor \log_p r \rfloor$, so $v_p(A^{-1}) \geq -b_p$. The initial residual has valuation at least $a_p - 1 = 2b_p + 1$ by (2). Lemma 2 completes the proof. $\square$

Corollary 4. *If $K \geq M_p$, then, for every $c = (c_1, \dots, c_r) \in \mathbb{Q}_p^r$, the system $\sum_i x_i^j = c_j$ ($1 \leq j \leq r$) has a nonsingular solution in $\mathbb{Q}_p^K$.*

Proof. At the nonsingular zero of Lemma 3, fix all but a set of r coordinates with invertible Jacobian. The p -adic inverse function theorem (or the contraction argument of Lemma 2) shows that all sufficiently small right-hand sides are attained. Choose a large integer e so that $(p^{ej}c_j)_{j=1}^r$ is such a right-hand side. A corresponding solution z gives $x_i = p^{-e}z_i$, and scaling does not change the rank of the Jacobian. $\square$

If $p \leq 4r^2$, then

$$M_p = p^{2b_p+2} \leq p^2 r^2 p^{2(r-1)/(p-1)} \leq 16r^6 4^{r-1} = 4r^6 4^r. \quad (3)$$

Here $v_p((r-1)!) \leq (r-1)/(p-1)$ and $p^{1/(p-1)} \leq 2$ were used.

3 Large primes: integral solutions for every target

Lemma 5. *If $p > 4r^2$ and $K \geq 4r^2$, then, for every $c \in \mathbb{Z}_p^r$, the system $\sum_i x_i^j = c_j$ ($1 \leq j \leq r$) has a nonsingular solution in $\mathbb{Z}_p^K$.*

Proof. First work modulo p , and fix the first r coordinates to be $0, 1, \dots, r-1$. Write $L = K - r$ and subtract their moments from c , leaving an arbitrary target $d \in \mathbb{F}_p^r$ for the L free coordinates. For each nonzero $\alpha = (\alpha_1, \dots, \alpha_r) \in \mathbb{F}_p^r$, the Weil bound for polynomial additive-character sums [3] gives

$$\left| \sum_{x \in \mathbb{F}_p} \exp \left(\frac{2\pi i}{p} \sum_{j=1}^r \alpha_j x^j \right) \right| \leq (r-1)\sqrt{p}.$$

The degree-one case has sum zero, and $p > r$ rules out the inseparable exceptions. Character orthogonality therefore bounds the number $N(d)$ of solutions for the free coordinates by

$$|N(d) - p^{L-r}| \leq (r-1)^L p^{L/2}.$$

The ratio of the error to p^{L-r} is at most

$$(r-1)^L p^{r-L/2} < (2r)^{2r} 2^{-L} < 1.$$

For the final inequality, $L \geq 4r^2 - r > 2r^2$ and $\log_2(2r) \leq r$ for $r \geq 2$ suffice. Thus a solution modulo p exists. Its fixed first r coordinates already give an invertible Vandermonde Jacobian minor modulo p . Ordinary multivariate Hensel lifting gives the claimed integral p -adic solution. $\square$

4 The global step

Proof of Theorem 1. Choose $K \geq Cr^6 4^r$, where C is an absolute constant large enough that $K \geq 4r^6 4^r$, $K \geq 3r^2 2^r - r$, and the real equal-weight quadrature theorem applies at degree $2r+2$ with K nodes. The real theorem for uniform measure on an interval ensures this for all $K \geq C_0(2r+2)^2$; see Gilboa–Peled [2]. A real rule of degree $2r+2$ has at least $r+2$ distinct support points: otherwise a nonzero squared polynomial of degree at most $2r+2$ vanishing on the support would contradict exactness. Consequently it has at least r distinct interior support points. The Jacobian for the first r moments at these points is invertible. The implicit function theorem therefore permits any endpoint nodes to move into $(0,1)$, with corrections to these r interior pivots, while retaining the first r moments. We obtain a fixed nonsingular real solution $u \in (0,1)^K$ to

$$\sum_i u_i^j = c_j, \quad c_j = K/(j+1), \quad 1 \leq j \leq r.$$

For each of the finitely many primes $p \leq 4r^2$, equations (3) and Corollary 4 give a $\mathbb{Q}_p$ solution. Choose an integer P_0 such that all coordinates of these finitely many solutions belong to $P_0^{-1}\mathbb{Z}_p$, and increase P_0 so that $c_j P_0^j \in \mathbb{Z}$ for every j . For every prime $p > 4r^2$, the target c is p -integral because $p > r+1$, and Lemma 5 gives a solution in $\mathbb{Z}_p^K$.

For every positive multiple P of P_0 , all the local solutions thus belong to $(P^{-1}\mathbb{Z}_p)^K$. For sufficiently large such P , the fixed real solution u belongs to $[P^{-1}, 1 - P^{-1}]^K$ and its separation quantity Δ_u from [1, Theorem 5.2(v)] is strictly positive. Theorem 5.2(iv),(v) gives a positive lower bound, independent of P , for the singular series and singular integral. The asymptotic formula in Theorem 5.2(i), whose error has a strictly smaller power of P , therefore gives a solution in $((0,1) \cap P^{-1}\mathbb{Z})^K$ for all sufficiently large multiples P of P_0 . $\square$

Scope. The theorem determines the exponential order of rational equal-weight interval quadrature size. It does not give exponentially small dice in the article’s whole-copy insertion construction, because the common denominators of the rational nodes are uncontrolled. In fact the general denominator tradeoff in `polynomial_denominator_designs.tex` implies that exponential-size rational rules must have exponentially large common denominator. The local and global arguments above have received independent line-by-line reviews from two collaborating agents; historical novelty still requires a separate literature investigation.

References

- [1] Zhen Cui, Jiacheng Xia, and Ziqing Xiang, *Rational designs*, Advances in Mathematics 352 (2019), 541–571, doi:10.1016/j.aim.2019.06.012.
- [2] Shoni Gilboa and Ron Peled, *Chebyshev-type quadratures for doubling weights*, Constructive Approximation 45 (2017), 193–216, arXiv:1507.01505.
- [3] André Weil, *On some exponential sums*, Proceedings of the National Academy of Sciences 34 (1948), 204–207, doi:10.1073/pnas.34.5.204.